SONGYUAN ZHANG

LAB: REALM Lab, MIT Building 31, 70 Vassar St, Cambridge, MA 02139, U.S.
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RESEARCH INTERESTS

Reinforcement Learning (Online/Offline), Robot Safety, Learning-based Control, Generative Policies, Embodied AI, Humanoid Whole-Body Control, Imitation Learning, Multi-Agent Systems, Control Theory

EDUCATION

Massachusetts Institute of Technology Sep. 2021 - May 2027 (expected)
Ph.D. Student, Department of Aeronautics and Astronautics *Advisor: Prof. Chuchu Fan*

Massachusetts Institute of Technology Aug. 2021 - Feb. 2024
Master of Science in Aeronautics and Astronautics *Advisor: Prof. Chuchu Fan*

Tsinghua University Sep. 2017 - Jul. 2021
B.E. in Tsien Excellence in Engineering Program (honors degree)
Advisors: Prof. Yanan Sui, Prof. Hao Wu, Prof. Gangtie Zheng

Stanford University Jun. 2020 - Jul. 2021
Remote Visiting Scholar in Computer Science *Advisor: Prof. Dorsa Sadigh*

University of California, Berkeley Jul. 2019 - Sep. 2019
Visiting Scholar in Mechanical Engineering *Advisor: Prof. Masayoshi Tomizuka*

RESEARCH AND PROFESSIONAL EXPERIENCE

Amazon Industrial Robotics Group Summer 2026
Applied Scientist II Intern *Project: Humanoid Whole-Body Safe Control*

- Developed FAITH (Feasibility-Aware Safety-Filtered RL for High-Dimensional Systems), a model-free RL framework that separates learned safety filtering from task-policy optimization, allowing the policy to maximize task performance without a competing safety penalty.
- Deployed FAITH on a 29-DoF Unitree G1 humanoid with onboard sensing. Demonstrated walking, push recovery, and falls away from protected regions using a single controller without explicit mode switching.

Massachusetts Institute of Technology Sep. 2021 - Now
Doctoral Research *Thesis: Structured Policy Learning for Reliable Robot Control*

- Formulated compositional stability certificates (NeurISS) and graph-based safety certificates (GCBF), enabling scalable, distributed control of large-scale multi-agent systems.
- Designed safe multi-agent control and RL algorithms (GCBF+, DGPPO, Def-MARL), scaling to 1,024 agents in simulation and demonstrating collision avoidance and collaborative tasks on Crazyflie drones.
- Developed support-constrained offline RL algorithms (ReFORM, LASER) that optimize flow-based generative policies in latent space, preserving learned action support to mitigate out-of-distribution errors in locomotion and manipulation tasks.

PUBLICATIONS

* *Equal contribution.*

[Preprint] Songyuan Zhang, Baljeet Singh, Sarthak Ranjeet Kaingade, Chuchu Fan, and Bryan Trinh. "FAITH: Feasibility-Aware Safety-Filtered RL for High-Dimensional Systems." *arXiv preprint arXiv:2610.12432*, 2026. [arXiv]

[NeurIPS'26] Songyuan Zhang, Oswin So, Eric Yang Yu, Matthew Cleaveland, Peter Crowley-Dolen, and Chuchu Fan. "LASER: Latent Space Adjoint Matching for Support-Constrained Entropy-Regularized Offline RL." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. [Website]

[CDC'26] Hao Wang, Max Cohen, Zeyuan Feng, Donggeon David Oh*, Oswin So*, Songyuan Zhang*, Javier Borquez, Chuchu Fan, Jaime Fernández Fisac, Haimin Hu, Jason J. Choi, and Somil Bansal. "Unified Overview of Safety Certificate Functions: Techniques from Hamilton–Jacobi Reachability and Control Barrier Functions." *65th IEEE Conference on Decision and Control (CDC)*, 2026. [arXiv]

[RA-L'26] Jaeyoun Choi, Oswin So, Songyuan Zhang, Cooper Taylor, and Chuchu Fan. "Safe and Scalable Multi-Drone Payload Transport via CBF-based Reinforcement Learning with Zero-Shot Sim-to-Real Transfer." *IEEE Robotics and Automation Letters (RA-L)*, 2026. [PDF]

[TIA'26] Qing Shen, Peng Zhang, Songyuan Zhang, Chuchu Fan, Yifan Zhou, and Yacov A. Shamash. "Hamiltonian Grid-Forming: Methodology and Applications for IBR-Rich Grid Operations." *IEEE Transactions on Industry Applications*, 2026. [PDF]

[ICRA'26] Akila Saravanan, Songyuan Zhang, Travis Manderson, Nicholas Roy, and Chuchu Fan. "Beyond Waypoints: Semantic-Centric Autonomy with Unreliable Maps Through Learned Abstractions." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [PDF]

[ICLR'26] Songyuan Zhang, Oswin So, H. M. Sabbir Ahmad, Eric Yang Yu, Matthew Cleaveland, Mitchell Black, and Chuchu Fan. "ReFORM: Reflected Flows for On-support Offline RL via Noise Manipulation." *International Conference on Learning Representations (ICLR)*, 2026. [Website]

[ICLR'26] Oswin So, Eric Yang Yu, Songyuan Zhang, Matthew Cleaveland, Mitchell Black, and Chuchu Fan. "Solving Parameter-Robust Avoid Problems with Unknown Feasibility using Reinforcement Learning." *International Conference on Learning Representations (ICLR)*, 2026. [Website]

[NeurIPS'25] H. M. Sabbir Ahmad, Ehsan Sabouni, Alexander Wasilkoff, Param Budhraja, Zijian Guo, Songyuan Zhang, Chuchu Fan, Christos Cassandras, and Wenchao Li. "HMARL-CBF – Hierarchical Multi-Agent Reinforcement Learning with Control Barrier Functions for Safety-Critical Autonomous Systems." *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [Website]

[RSS'25 (Outstanding Student Paper Award)] Songyuan Zhang*, Oswin So*, Mitchell Black, Zachary Serlin, and Chuchu Fan. "Solving Multi-Agent Safe Optimal Control with Distributed Epigraph Form MARL." *Proceedings of Robotics: Science and Systems (RSS)*, 2025. [Website]

[ICLR'25] Songyuan Zhang, Oswin So, Mitchell Black, and Chuchu Fan. "Discrete GCBF Proximal Policy Optimization for Multi-agent Safe Optimal Control." *International Conference on Learning Representations (ICLR)*, 2025. [Website]

[T-RO'25] Songyuan Zhang*, Oswin So*, Kunal Garg, and Chuchu Fan. "GCBF+: A Neural Graph Control Barrier Function Framework for Distributed Safe Multi-Agent Control." *IEEE Transactions on Robotics*, 2025. [Website]

[L4DC'24] Songyuan Zhang and Chuchu Fan. "Learning to stabilize high-dimensional unknown systems using Lyapunov-guided exploration." *6th Annual Learning for Dynamics & Control Conference (L4DC)*, 2024. [Website]

[**ARC'24**] Kunal Garg, Songyuan Zhang, Oswin So, Charles Dawson, and Chuchu Fan. "Learning Safe Control for Multi-Robot Systems: Methods, Verification, and Open Challenges." *Annual Reviews in Control*, 2024. [arXiv]

[**CoRL'23**] Songyuan Zhang, Kunal Garg, and Chuchu Fan. "Neural Graph Control Barrier Functions Guided Distributed Collision-avoidance Multi-agent Control." *7th Conference on Robot Learning (CoRL)*, 2023. [Website]

[**SG'23**] Lizhi Wang, Songyuan Zhang, Yifan Zhou, Chuchu Fan, Peng Zhang, and Yacov A. Shamash. "Physics-Informed, Safety and Stability Certified Neural Control for Uncertain Networked Microgrids." *IEEE Transactions on Smart Grid*, 2023. [PDF]

[**PESGM'23**] Lizhi Wang, Songyuan Zhang, Yifan Zhou, Chuchu Fan, Peng Zhang, and Yacov A. Shamash. "Learning-Based, Safety and Stability-Certified Microgrid Control." *2023 IEEE Power & Energy Society General Meeting (PESGM)*, 2023. [PDF]

[**L4DC'23 (ORAL)**] Songyuan Zhang, Yumeng Xiu, Guannan Qu, and Chuchu Fan. "Compositional Neural Certificates for Networked Dynamical Systems." *5th Annual Learning for Dynamics & Control Conference (L4DC)*, 2023. [Website]

[**NeurIPS'21**] Songyuan Zhang*, Zhangjie Cao*, Dorsa Sadigh, and Yanan Sui. "Confidence-Aware Imitation Learning from Demonstrations with Varying Optimality." *Conference on Neural Information Processing Systems*, 2021. [Website]

HONORS AND AWARDS

2025-2026 Schwarzman College of Computing Amazon AI Research Innovation Fellowship	2025
Outstanding Student Paper Award, Robotics: Science and Systems (RSS)	2025
Outstanding Paper Award Finalist, Robotics: Science and Systems (RSS)	2025
Outstanding Graduate of Tsinghua University	2021
Outstanding Graduate of Beijing	2021
Outstanding Graduate of Scientific Research at TEEP	2021
Outstanding Graduate of Creativity at TEEP	2021
Honor Degree of Tsien Class, Tsinghua University	2021
Tsinghua Scholarship for Technological Innovation	2020
National Scholarship, China	2019
First Prize in Excellent Student Research Training Contest of Tsinghua University	2019
First Prize in "Challenge Cup" Science and Technology Competition of Tsinghua University	2019

ACADEMIC & PROFESSIONAL SERVICE

Workshop Organizing Committee Member: Workshop on Safety Analysis for Autonomous System: Practical Perspectives and Recent Advances, 65th IEEE Conference on Decision and Control (CDC), 2026

Panel Moderator: Robotics × Physical AI Founder-Investor Summit – Academia panel on frontier research, world models, and safe autonomy (July 2026)

Journal Reviewer:
IEEE Transactions on Robotics (T-RO), IEEE Transactions on Automatic Control (T-AC), Automatica, Annual Reviews in Control (ARC), IEEE Open Journal of Control Systems (OJ-CSYS), IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Cybernetics (T-CYB), IEEE Transactions on Control Systems Technology (TCST)

Conference Reviewer:
NeurIPS, ICLR, ICML, RSS, CoRL, ICRA, L4DC, IROS, AAAI, AISTATS, CDC, ACC, NeuS, NFM, DARS, RV, TACAS, ANTS, CCTA

Teaching Assistant: X-idea, Tsinghua University (Spring 2021)

LEADERSHIP & MEMBERSHIP

Chairman of 13th *Spark* Innovative Talents Training Program at Tsinghua University

Vice-monitor of Tsien Class 2017

Member of the Second Team of Middle and Long Distance Running at Tsinghua University

SELECTED PRESS COVERAGE

MIT engineers help multirobot systems stay in the safety zone, *MIT News*, 2025. [Website]

Nine MIT PhD students named Amazon AI Fellows, *MIT Schwarzman College of Computing*, 2025. [Website]

HOBBIES

Travelling, Hiking, Photography, Badminton, Running, Skiing